Atomic Fee Enforcement in Decentralized Perpetual Trading

A Transaction-Layer Approach Using Elliptic Curve Cryptography

BITS F452: BLOCKCHAIN TECHNOLOGY

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Motivation

The Fee Collection Vulnerability

The Problem

Modern decentralized exchanges separate trade execution and fee collection into independent transactions, creating a vulnerability where trades may succeed while fee transactions fail.

This breaks atomic transaction guarantees, allowing users to exploit protocol logic and bypass cryptographically enforced fee payment mechanisms.

Why This Matters

This architectural gap undermines DEX sustainability, enabling revenue loss, front-running, and manipulation opportunities. Without cryptographic atomicity, trustless execution weakens, violating blockchain consistency principles and exposing economic and security risks that threaten long-term protocol reliability and decentralized financial ecosystem integrity.

Objectives

1. Atomic Fee Bundling:

Design a transaction architecture that cryptographically binds fee settlement with trade execution, ensuring inseparable, all-or-nothing atomic transaction enforcement at the protocol level.

2. Signing Layer Cryptographic Integration:

Embed fee instructions directly into the Ed25519 signing process, ensuring a single cryptographic authorization simultaneously validates trade execution and fee payment.

3. Transaction Integrity Verification:

Utilize SHA-256 hashing to preserve transaction integrity, guaranteeing that modifications to trade or fee parameters invalidate signatures and prevent tampering.

4. Comprehensive Trade Primitive Support:

Extend atomic fee guarantees across market, limit, and stop-loss orders while maintaining deterministic execution and consistent fee enforcement logic.

5. Blockchain-Native Atomic Execution Architecture:

Leverage Solana's versioned transaction model and Proof-of-History consensus to enforce ordered instruction execution and eliminate race-condition vulnerabilities.

6. Security and Attack Resistance Validation:

Evaluate system resilience against replay attacks, signature forgery, fee bypass attempts, and transaction manipulation through controlled adversarial testing scenarios.

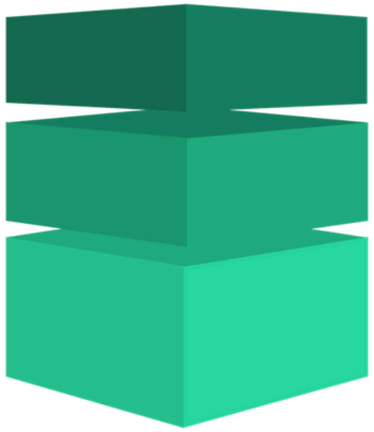
Three-Phase Plan to Achieve Our Objectives

Research Phase

Define architecture & model

Validation Phase

Test cryptographic fee interception



Development Phase

Implement layered blockchain modules

Phase 1: Research & Architectural Design

Study Solana transaction processing, signing workflows, and DEX execution models to design a cryptographically enforced atomic fee interception architecture and identify secure instruction injection points.

Phase 2: System Development & Cryptographic Integration

Implement the transaction interceptor, fee calculation engine, and Ed25519 signature integration while ensuring compatibility with trading order types and preserving blockchain transaction validity.

Phase 3: Validation, Security Analysis & Performance Evaluation

Test atomic execution guarantees through adversarial scenarios, validate resistance against replay and bypass attacks, and evaluate performance overhead to confirm scalability and cryptographic correctness.

Expected Outcomes

1	**Functional Atomic Fee Enforcement System** Develop a DEX-compatible trading system enabling atomic, inseparable execution of trades and fee settlements.
2	**Verified Cryptographic Transaction Authorization** Implement Ed25519 signatures and SHA-256 hashing to ensure secure transaction authentication and integrity.
3	**Demonstrated Security and Attack Resistance** Validate resistance against replay attacks, signature forgery, fee bypass attempts, and transaction manipulation.
4	**Performance and Scalability Validation** Benchmark system overhead, confirming minimal latency impact and preserved throughput for high-frequency decentralized trading.

References

Blockchain Architecture

• Solana Documentation: Transaction structure, instruction formats, and account execution model (docs.solana.com)
• Solana Programming Model: Runtime specifications and security considerations

Cryptographic Standards

• RFC 8032: Ed25519 digital signature specification and security properties
• FIPS 180-4: SHA-256 secure hash standard for transaction integrity

Applied Cryptography

• Menezes, van Oorschot, Vanstone: Handbook of Applied Cryptography (Digital Signatures)
• Boneh, Shoup: A Graduate Course in Applied Cryptography (Signature Schemes and ECC)